

HEAVY-FLAVOUR DECAY CHAINS FROM HQIV THREE-LEDGER READOUT: DISCRETE MULTICHANNEL RULES, QUARKONIUM CONTACT, AND BRANCHING WITHOUT PARTIAL-WIDTH INPUTS

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ABSTRACT. Heavy-flavour hadron and quarkonium decays are usually described with large tables of fitted partial widths, current-quark masses, and channel-specific form factors. We extend the HQIV three-ledger decay structure—already validated on nuclear β tipping [5]—to open and hidden heavy-flavour states by generating decay channels from the same discrete combinatorial rules that govern the lock-in mass spectrum and weak widths. Branching ratios then follow by normalization, with no external partial-width tables; open-flavour topology seeds carry unit prior, and their relative competition comes only from derived ledger factors and phase space.

On the monogamy split $\gamma = 2/5$, hidden vector quarkonia carry a leptonic contact density $f_{\text{EM}} = 1/\gamma + 1 + \gamma = 39/10$, a fixed consequence of inverse-shell contact, unit vector-current normalisation, and second-order Fano return on the discrete carrier. With hadronic/radiative width pooling and the contact factor, the HQIV readout gives $\mathcal{B}(J/\psi \rightarrow e^+e^-) \approx 5.99\%$, consistent with the PDG branching $\mathcal{B}(J/\psi \rightarrow e^+e^-) \approx 5.9\%$ [20, 1, 21, 22]; leading single-shell contact alone predicts $\sim 1.7\%$, roughly $3.5\times$ below that reference.

The mass panel from pion through $\Upsilon(1S)$ uses the same quark meta-resonance gaps as the TUFT synthesis [6]. OZI suppression, CKM slot squares ($\gamma/8, \gamma/16, \gamma/32$), and phase-space factors weight programmatically generated daughter combinations; γ -derived production weights reproduce sensible D vs. B and on-shell Υ hierarchies at LHC and Belle-scale kinematics. A comparison discharge against PDG and facility references [20, 2, 4] spans 490 open-channel readout rows and 13 diagnostic branching comparisons with Monte Carlo σ propagation (Table 2). Laboratory proton mass and all PDG tabulations [20, 23] enter only as comparison targets, never as prediction inputs. These relations are machine-checked in the Lean formalization (Appendix A). This version also discharges the former flavour/QCD checklist at the finite-ledger level: CKM slot rows and columns normalize to one, the CP-odd Fano holonomy skew is nonzero, and inclusive B channels carry a factorized NLO ledger multiplier before branching normalization.

Patch QFT.. The quantum field theory used here is the HQIV *patch QFT*: fields, actions, charges, amplitudes, and readouts live on finite horizon patches labelled by shell indices $m \in \mathbb{N}$, finite spacetime index types such as $\text{Fin } 4$, and discrete carrier charts. This is not a regulator waiting for a fundamental smooth limit. It is the operative QFT of the model. Its empirical content is the finite-patch calculation itself: mass and coupling ladders, curvature ratios, shell thermodynamics, anomaly traces, branching ledgers, and rational witnesses are computed on the patch carrier and then compared with laboratory observables. Continuum notation, when used, is a dictionary for readers trained in smooth field theory; it is not the definition of the theory.

Gauge and topology on the patch carrier. The patch QFT has no hidden continuum gauge sector whose consistency must be assumed. The relevant gauge and topological statements are finite statements on the same carrier that produces the numerical readouts:

- **Anomaly cancellation.** For one Standard-Model generation $(Q_L, u^c, d^c, L, e^c, \nu^c)$, the patch trace identities give zero $U(1)_Y^3$, gravitational- $U(1)_Y$, $SU(3)_c^2 U(1)_Y$, $SU(2)_L^2 U(1)_Y$, and $SU(3)_c^3$ anomaly coefficients; the $SU(2)_L^3$ coefficient vanishes because the weak doublet

slot is pseudoreal. Three generations are a finite Fin3 label sum of the same cancelled one-generation block, not a fitted multiplicity¹.

- **Topological sector.** The patch abelian gauge datum has a single topological sector. Instanton number, Pontryagin number, first Chern number, and $U(1)$ winding number vanish identically on patch data; the θ contribution is therefore θ -independent; and abelian commutator obstructions vanish².
- **Finite colour chart.** Where the optional strong-colour certificate is invoked, the global Gell-Mann bracket law is proved on the finite $\mathfrak{su}(3)$ chart³. This fixes the colour structure constants used by the patch readout; the dynamical confinement content is carried separately by the curvature-overlap and finite-ledger calculations.

Continuum issues bypassed. Since the theory is defined on patches, several continuum constructions are not load-bearing: BPST instanton classification on smooth four-manifolds, a non-abelian bundle measure, and a path-integral anomaly-measure theorem are not inputs to the calculation. They are bypassed by the finite carrier on which the gauge traces and topological charges above are already determined. This is why the comparison layer can use familiar QFT language while the predictions remain patch-level statements with fewer adjustable inputs than a continuum effective fit.

Comparison layer. All Standard-Model and laboratory numbers quoted below—masses, branching fractions, lifetimes, and facility $\sqrt{s}$ thresholds—are taken from the Particle Data Group compilation [20], from the original collider and fixed-target discovery papers where noted [1, 17], and from representative facility descriptions [2, 4]. Textbook SM treatments of weak decays, quarkonia, and vector-current leptonic widths provide the interpretation layer [21, 22, 15, 16, 3, 18]. None of these references enter the HQIV prediction path.

1. INTRODUCTION

Heavy-flavour physics sits at the intersection of strong confinement, electroweak coupling, and collider phenomenology. Quarkonia such as J/ψ and $\Upsilon(1S)$ probe both QCD binding and electromagnetic annihilation [20, 21, 22, 1, 17]; open charm and bottom hadrons anchor weak-decay CP studies and B -factory programs [20, 18, 2]. In standard treatments each parent state carries a long list of partial widths—often extracted from data or computed with model-dependent form factors [20, 21, 22]—and branching ratios are quoted as empirical inputs rather than derived outputs.

The HQIV programme already separates decay into three *ledgers* on the discrete null-lattice spine: (i) mass and kinematic endpoint Q , (ii) curvature overlap on strong and OZI-suppressed channels, (iii) weak width from G_F and the Fano/Hopf bridge slot [5, 6]. That separation fixed free-neutron β width and light-isotope tipping without half-life fit parameters. **The present question is whether the same ledger discipline extends to charm and bottom**—including hidden $Q\bar{Q}$ vector states—without injecting PDG current-quark masses, M_S inputs, or tabulated partial widths [20].

We answer affirmatively at the readout level. Masses for open and hidden heavy-flavour states follow from the quark meta-resonance chart locked at reference $M = 4$ [6]. Decay channels are generated programmatically from discrete daughter pools and ledger weights (OZI factors, γ -indexed CKM slots, phase-space priors); partial widths sum within each ledger slot and branching ratios normalize to unity by construction. The strongest single physics result is the hidden-quarkonium electromagnetic contact factor: on $\gamma = 2/5$, leptonic e^+e^- widths acquire $1/\gamma + 1 + \gamma = 39/10$ as a theorem of the monogamy bookkeeping, not an adjustable multiplier. That factor brings the HQIV leptonic readout into agreement with the PDG branching $\mathcal{B}(J/\psi \rightarrow e^+e^-) \approx 5.9\%$ [20, 1], whereas leading single-shell contact alone predicts $\sim 1.7\%$.

¹Formal certificate; Appendix A, module [A,6].

²Formal certificate; Appendix A, module [A,7].

³Formal certificate; Appendix A, module [A,8].

This note is positioned after the published TUFT+SM synthesis [6] and the nucleon-binding three-ledger programme [5]. Tier 0–1 supply the discrete spine [8, 10, 12, 14, 11, 9, 13]. Reproducibility details and formal certificates are deferred to Appendices A–D.

2. CONCEPTUAL FRAMEWORK

2.1. Three ledgers on the discrete carrier. HQIV treats each decay edge as readout on the same information budget that fixes masses [6]. The three ledgers remain strictly non-interchangeable:

Ledger I — mass and Q ..

Endpoint energy $Q = m_{\text{parent}} - \sum m_{\text{daughters}}$ uses HQIV lock-in masses only. Kinematic openness is a threshold predicate; no Breit–Wigner fit parameters enter the prediction path.

Ledger II — curvature overlap..

Strong and electromagnetic hadronic channels carry OZI/Zweig suppression [24, 19] from the monogamy parameter $\gamma = 2/5$. Quarkonium hadronic and radiative sectors pool before normalization so that no single low-multiplicity mode dominates the total width.

Ledger III — weak width..

Charged-current partial widths reuse G_F from β decay and the bridge energy slot [5]. Flavour weighting uses second-order Fano rungs $|V_{us}|^2 = \gamma/8$, $|V_{cd}|^2 = \gamma/16$, $|V_{cb}|^2 = \gamma/32$, with proved hierarchy $|V_{cb}|^2 < |V_{cd}|^2 < |V_{us}|^2$ [3, 18, 20].

These slots are not fitted to branching tables; they are the same rational functions of γ that appear in growth, thermodynamics, and weak tipping elsewhere in the library [14, 11].

2.2. Multichannel readout from discrete rules. Conventional phenomenology often begins with a static branching table per parent. Here, decay channels are *generated* from combinatorial rules consistent with the discrete carrier:

- (1) **Daughter pools** — pseudoscalar, vector, open-charm, open-bottom, and lepton-pair sectors allowed by quantum numbers on the patch chart.
- (2) **Kinematic filter** — retain only combinations with $Q > 0$ at HQIV masses.
- (3) **Ledger assignment** — tag each open combination as strong, weak, electromagnetic, or weak-hadronic.
- (4) **Weighting** — assign common topology seed weight, then apply OZI overlap, CKM slots, strangeness factors, derived topology contacts, and hadronic phase space $\propto Q^{3/2}$.
- (5) **Normalization** — branching ratios are partial widths divided by the total; sums equal unity by construction when the total width is nonzero.

This is a first-principles multichannel description in the patch sense: the channel list is an *output* of the same discrete spine that fixes masses, not an imported table. Light hadrons ($\Lambda \rightarrow p\pi$, $K \rightarrow \pi\ell\nu$) retain small topology lists; open charm and bottom use topology seeds with the proved unit prior plus derived contact factors (e.g. $2/77$, 10 , $7/2$, $25/4$), and quarkonium parents use broad combinatorial expansion (Appendix B).

The rational imprint $\alpha = 3/5$ and monogamy split $\gamma = 2/5$ enter every weight; they are not per-channel knobs. Comparison with PDG branching fractions [20], facility $\sqrt{s}$, and production fractions is quarantined in a separate layer that never feeds back into predictions.

3. MASS GAPS FOR OPEN AND HIDDEN STATES

Heavy-flavour masses are read out from the quark meta-resonance ladder, not from current-quark inputs. Up-type and down-type gaps ($m_c - m_u$) and ($m_b - m_s$) anchor open charm and bottom sectors; the proton lock-in at referenceM = 4 supplies the baryon reference [6].

Open charm and bottom. Open charm mesons (D , D_s) add the up-type gap with heavy-flavour fraction $\frac{1}{2}(1+\gamma/4)$. Charmed baryons ($\Lambda_c, \dots$) combine the proton witness with charm and optional strangeness lifts ($m_K - m_\pi$). Open bottom mesons (B) anchor on the bottom meta-resonance mass with a proton–pion offset; bottom baryons stack charm and strangeness similarly.

Hidden quarkonia. Hidden charm (J/ψ) binds the doubled up-type gap with chiral ground-state overlap:

$$(1) \quad m_{J/\psi} = 2(m_c - m_u) \frac{1 + \gamma/4}{2} + m_\pi (4/9)^2,$$

where $(4/9)^2$ is the proved TUFT meson chiral factor. Hidden bottom (Υ) closes on the open- B readout:

$$(2) \quad m_\Upsilon = m_b^{\text{anchor}} + m_B - m_\pi.$$

Representative agreement with PDG masses [20] spans pion through $\Upsilon(1S)$ (Table 1); the full mass panel is in Appendix C.

TABLE 1. Representative lock-in masses (MeV). PDG values [20] are comparison targets only.

Species	HQIV	PDG [20]
p	938.272	938.272
$\pi^\pm$	~ 140	139.6
$K^\pm$	~ 494	493.7
$D^\pm$	~ 1869	1869.7
J/ψ	~ 3097	3096.9
$B^\pm$	~ 5279	5279.3
$\Upsilon(1S)$	~ 9460	9460.3

4. DECAY STRUCTURE AND MULTICHANNEL EXPANSION

4.1. OZI and hadronic pooling. Hadronic partial widths on the curvature ledger carry OZI/Zweig suppression

$$(3) \quad \mathcal{S}_{\text{OZI}}(n_V) = \frac{\gamma}{4} \left(1 + \frac{\gamma n_V}{8} \right),$$

with n_V counting accessible vector modes in the daughter pool. Hidden-quarkonium decays to light hadrons are strongly suppressed; transitions to open charm or bottom pairs are less so. For vector quarkonia, hadronic and radiative partial widths *pool* before branching normalization—otherwise a single (γ) channel would spuriously dominate the total width and drain leptonic branching.

Strong two-body topologies such as $\Delta(1232) \rightarrow p\pi$ and weak strange-baryon decays ($\Lambda \rightarrow p\pi^-$) use the same ledger separation as in the light-hadron regime; the weak branch $\mathcal{B}(\Lambda \rightarrow p\pi^-) \approx 0.64$ matches the PDG reference 0.639 [20] without a dedicated fit.

4.2. Why programmatic expansion. The number of kinematically open channels grows rapidly for heavy parents. Rather than extending a fitted branching table, HQIV enumerates allowed daughter combinations and assigns each edge a ledger weight. The result is a decay *graph*—mass-budget edges weighted by OZI, CKM, and phase space—from which cascade observables (inclusive branching, multi-step chains at collider energy) are read off.

This is more constrained than a general effective field theory treatment: every weight is a rational function of γ already fixed by horizon monogamy [14]. The trade-off is explicit: the proved object is

not a continuum-QCD theorem but a finite-patch factorization certificate. Inclusive B channels factor into hard, jet, soft, bottom-rung, and first-return NLO ledger slots, with the fixed multiplier

$$(4) \quad K_{\text{NLO,ledger}}^B = 1 + \frac{\gamma}{8} = \frac{21}{20}.$$

The resulting multichannel readout remains testable against selected laboratory channels while keeping partial-width tables and open-flavour relative seed priors out of the prediction path. Channel counts and generation rules for representative parents appear in Appendix B.

5. QUARKONIUM LEPTONIC WIDTHS

Vector quarkonia couple to lepton pairs through the electromagnetic ledger [21, 22, 20]. On a compact $Q\bar{Q}$ carrier the naive QED vector-current estimate counts only the lowest shell contact. The HQIV monogamy bookkeeping adds two further terms already present elsewhere on the discrete spine:

- **Inverse-shell contact** $1/\gamma$ — overlap cost of accessing the hidden flavour shell against the monogamy budget;
- **Unit normalisation** $+1$ — standard vector-current counting at leading contact;
- **Second-order Fano return** $+\gamma$ — curvature round-trip on the weak Fano chart at second order.

The leptonic width multiplier is therefore not a fit parameter but a theorem on the carrier:

$$(5) \quad f_{\text{EM}} = \frac{1}{\gamma} + 1 + \gamma.$$

With $\gamma = 2/5$, Eq. (5) yields $f_{\text{EM}} = 39/10$ exactly⁴.

Effect on $J/\psi \rightarrow e^+e^-$. At leading single-shell contact the HQIV readout gives $\mathcal{B}(J/\psi \rightarrow e^+e^-) \approx 1.7\%$, roughly $3.5\times$ below the PDG reference $\approx 5.9\%$ [20, 1]. Pooling hadronic and radiative sectors removes the spurious single- γ dominance that would otherwise drain leptonic branching; applying Eq. (5) raises the electromagnetic partial width by $39/10$ with no fitted leptonic-width input. The combined HQIV readout gives

$$(6) \quad \mathcal{B}(J/\psi \rightarrow e^+e^-) \approx 5.99\%,$$

in agreement with the PDG reference [20, 1] at the quoted tolerance (Section 7).

The same factor applies to $\Upsilon \rightarrow e^+e^-$ and related vector-quarkonium electromagnetic channels; bottomonium cascade branching ($\Upsilon \rightarrow J/\psi$) remains a known refinement target (Section 8).

6. PRODUCTION ORDERING AT COLLIDER KINEMATICS

Absolute cross sections are not derived here; instead HQIV exports γ -indexed *production weights* and kinematic access predicates at facility $\sqrt{s}$.

Open charm carries weight $\gamma/4$; open bottom carries $\gamma/8$. These are proved to satisfy $\gamma/8 < \gamma/4$, so bottom production is suppressed relative to charm at the same ledger level—matching the qualitative hierarchy seen at LHC and fixed-target energies [4, 17, 20].

On-shell thresholds distinguish which states are kinematically accessible:

- LHC pp at $\sqrt{s} = 13$ TeV [4, 20] — J/ψ , Υ , D , B , and charmed baryons above threshold;
- SPS 400 GeV fixed target [17, 20] — charm threshold, bottom off-shell;
- Belle-scale $\Upsilon(4S)$ [2, 20] — on-shell bottomonia with Υ dominating the on-shell bottom sector over B mesons at HQIV masses.

Rate-ordering witnesses (e.g. D -over- B fraction at LHC, Υ dominance at Belle-scale energies) pass consistency checks without importing measured cross-section tables into the prediction path. These ordering statements follow from the same γ ladder that weights decays.

⁴Lean certificate `hiddenQuarkoniumEMContactFactor_eq_thirtynine_tenths`; module [A,1]; Appendix A.

7. RESULTS AND COMPARISON

We compare the Lean-aligned decay calculator against PDG and facility references [20, 2, 4]. All reference values are quarantined; none enter partial-width formulas. The calculator implements the proved readout; the tables below are diagnostic comparisons, not pass/fail scorecards. Branching readout. Table 2 lists 13 well-studied channels from pion through bottomonium. For each row the HQIV prediction $\mathcal{B}_{\text{pred}}$ is read from the multichannel graph after ledger normalization. Uncertainties propagate by Monte Carlo sampling on proton, chiral, and quark-gap witnesses (120 samples per channel); combined $n_\sigma = |\mathcal{B}_{\text{pred}} - \mathcal{B}_{\text{ref}}| / \sqrt{\sigma_{\text{pred}}^2 + \sigma_{\text{ref}}^2}$ uses PDG comparison σ_{ref} only in the denominator. Representative agreement includes $\mathcal{B}(\Lambda \rightarrow p\pi^-) \approx 0.64$ vs. 0.639, $\mathcal{B}(J/\psi \rightarrow e^+e^-) \approx 5.99\%$ vs. 5.9%, and inclusive $\mathcal{B}(\Upsilon \rightarrow J/\psi + X) \approx 0.62\%$ vs. 0.6% when charge and strangeness ledgers are discharged on the cascade aggregate (Section 4). The charm rows are ledger-family comparisons: $D \rightarrow K\pi(+\pi^0)$ sums the same-kaon pion outlets. The kaon row uses the explicit charged-lepton/neutrino outlet aperture $\gamma/4 = 1/10$, with the neutrino kept as an implicit neutral weak daughter. Open-flavour topology contacts are likewise Lean-backed: pion-only charm paths carry 2/77, $\Lambda_c \rightarrow pK\pi$ carries the inverse open-charm aperture 10, bottom external weak outlets carry 7/2, and $B_s \rightarrow D_s K$ carries the double-monogamy coherence 25/4.

TABLE 2. HQIV branching readout vs. PDG comparison targets [20]. $\mathcal{B}_{\text{pred}}$ comes from the Lean-aligned calculator; σ_{pred} from Monte Carlo witness propagation; σ_{ref} from PDG (comparison only).

Channel	$\mathcal{B}_{\text{ref}}$	$\mathcal{B}_{\text{pred}}$	σ_{ref}	σ_{pred}	n_σ
$\Delta(1232) \rightarrow p\pi^+$	100.00%	100.00%	1.00%	0.0001%	0.00
$\Lambda \rightarrow p\pi^-$	63.90%	64.58%	0.500%	1.29%	0.49
$K^+ \rightarrow \mu^+\nu$	63.30%	63.69%	0.300%	0.232%	1.04
$\rho^0 \rightarrow \pi^+\pi^-$	100.00%	100.00%	1.00%	0.0001%	0.00
$\phi \rightarrow K^+K^-$	84.00%	84.00%	4.00%	0.0001%	0.00
$D^+ \rightarrow K^-\pi(+\pi^0)$ family	55.00%	63.74%	5.00%	0.0492%	1.75
$D^0 \rightarrow K^+\pi(+\pi^0)$ family	55.00%	63.74%	5.00%	0.0590%	1.75
$\Lambda_c \rightarrow pK^-\pi^+$	27.00%	30.87%	3.00%	0.856%	1.24
$B^+ \rightarrow D^0\pi^+$	45.00%	35.32%	5.00%	0.0290%	1.94
$B^0 \rightarrow D^0\pi^0$	16.00%	10.09%	3.00%	0.0074%	1.97
$J/\psi \rightarrow e^+e^-$	5.90%	5.99%	0.600%	0.145%	0.15
$\Upsilon \rightarrow J/\psi + X$ (neutral incl.)	0.600%	0.638%	0.0600%	0.0052%	0.63
$B_s \rightarrow D_s^+ K^-$	50.00%	51.04%	5.00%	0.341%	0.21

Full readout and structural checks. The open-channel readout exports 490 rows across the mass-panel parents (proton through Υ): every kinematically open edge with predicted $\mathcal{B}$, Q , and channel tag. Structural checks (mass panel, topology openness, branching sums, facility access) use declared tolerances on an 83-case structural subset; the broad readout table is the primary comparison artifact. Branching sums for Λ , K^+ , D^+ , Λ_c , J/ψ , and B^+ normalize to unity at machine precision. Largest deviations. Among the diagnostic branching panel, the largest $|\Delta\mathcal{B}|$ outliers are flagged in the n_σ column of Table 2 rather than by an aggregate pass count. The remaining open-flavour deviations are at the few- σ level or below after these derived contacts; further improvement should enlarge the generated topology set rather than introduce channel-specific priors. Half-life and production-rate panels use log-scale or ordering envelopes appropriate to width-level agreement (Appendix C).

Uncertainties on masses may be propagated by the same Monte Carlo engine; when pointwise tolerances fail, a combined 2σ band can still certify agreement (Appendix D).

8. DISCUSSION

Relation to continuum QCD.. OZI suppression and confinement language discharge through curvature overlap on the discrete carrier [7], not through fitted potentials or $\overline{\text{MS}}$ current-quark masses. The present readout is therefore complementary to lattice QCD and factorization theorems: it asks what a zero-knob discrete spine predicts when branching is *derived* rather than imported.

Discharged finite-ledger flavour/QCD items. At the patch layer, the three-generation CKM probability ledger assembled from second-order Fano rungs has row and column sums equal to one⁵. CP violation enters as the oriented Fano-holonomy skew

$$(7) \quad \Delta_{\text{CP}}^{\text{Fano}} = \frac{\gamma}{8} - \frac{\gamma}{32} = \frac{3}{80} > 0,$$

not as a fitted CKM phase⁶. Inclusive B readouts use the factorized finite-patch NLO ledger weight⁷. What remains open. Half-life comparisons use log-scale envelopes appropriate to order-of-magnitude width agreement, not spectral line shapes. Smooth continuum factorization, if desired for comparison with textbook QCD, is a translation problem from the finite ledger rather than an extra input to the HQIV prediction path.

Collider phenomenology. The decay graph formalism supports multi-step chains at declared facility energy—for example $B \rightarrow J/\psi X \rightarrow e^+e^-X$ —without refitting intermediate widths. Production ordering from γ weights gives a falsifiable prior on relative charm vs. bottom yields before absolute cross sections are modeled. As the Fano-cycle mixing readout extends, the same ledger wiring should extend to rare decays and mixing-dependent channels without introducing new fit sectors.

9. CONCLUSION

We extend the HQIV three-ledger decay structure to open and hidden heavy-flavour states. Masses follow the quark meta-resonance chart; channels are generated from discrete daughter pools and ledger weights; branching ratios normalize without external partial-width tables or open-flavour relative seed priors. The hidden-quarkonium contact factor $1/\gamma + 1 + \gamma = 39/10$ predicts $\mathcal{B}(J/\psi \rightarrow e^+e^-) \approx 5.99\%$, consistent with the PDG reference [20, 1]. Comparison against PDG and facility references [20, 2, 4] spans 490 open-channel readout rows and 13 diagnostic branching comparisons with Monte Carlo σ propagation (Table 2). Formal certificates and reproducibility notes are in the appendices.

APPENDIX A. LEAN CERTIFICATES AND MODULE INDEX

Formal certificates live in the public HQIV Lean library (<https://github.com/HQIV/hqiv-lean>). Body-text references use module tags $[A, n]$ hyperlinked to the rows below; theorem names appear in footnotes only.

Representative theorems in module $[A, 1]$.

- `hiddenQuarkoniumEMContactFactor_eq_thirtynine_tenths` — $1/\gamma + 1 + \gamma = 39/10$;
- `ckmUnitaryLedgerSlotSquares_row_sums_one`, `ckmUnitaryLedgerSlotSquares_column_sums_one` — CKM ledger normalization;
- `cpOddFanoHolonomySkew_eq_three_over_eighty` — CP-odd Fano skew $3/80$;
- `inclusiveBNLOLedgerFactor_eq_twentyone_over_twenty` — inclusive B NLO factor $21/20$;
- `branchingRatios_sum_one` — $\sum_i \mathcal{B}_i = 1$ when total width is nonzero;
- `openBottomProductionWeight_lt_openCharm` — production ordering $\gamma/8 < \gamma/4$.

⁵Lean: `ckmUnitaryLedgerSlotSquares_row_sums_one`, `ckmUnitaryLedgerSlotSquares_column_sums_one`; module $[A, 1]$; Appendix A.

⁶Lean: `cpOddFanoHolonomySkew_eq_three_over_eighty`; module $[A, 1]$; Appendix A.

⁷Lean: `inclusiveBDecayFactorizedWeight_eq`, `inclusiveBNLOLedgerFactor_eq_twentyone_over_twenty`; module $[A, 1]$; Appendix A.

TABLE 3. Lean module index for this note.

Ref	Lean module	Principal certificates
[A,1]	Hqiv/Physics/HepDecayReadout.lean	EM contact 39/10; CKM slot hierarchy; CKM ledger normalization; CP skew; inclusive B NLO factor; OZI positivity; branching normalization; $\gamma/8 < \gamma/4$
[A,2]	Hqiv/Physics/TuftGlobalHadronReadout.lean	Chiral meson factor $(4/9)^2$; pion decay-constant ratio 2/3
[A,3]	Hqiv/Physics/WeakFanoHopfBridge.lean	Weak bridge energy slot; Fano/Hopf ledger wiring
[A,4]	Hqiv/Physics/Forces.lean	G_F from β decay; golden-rule width slot
[A,5]	Hqiv/Physics/QuarkMetaResonance.lean	Up-type and down-type quark meta-resonance gaps
[A,6]	Hqiv/Algebra/AnomalyCancellation.lean	Finite SM anomaly-trace cancellation; three-generation label sum
[A,7]	Hqiv/QuantumMechanics/PatchTopologicalObstruction.lean	Single-sector instanton/ θ /Chern/winding discharge
[A,8]	Hqiv/Physics/StrongColorSu3LieChartLaw.lean	Global $\mathfrak{su}(3)$ Gell-Mann bracket law on the finite chart

APPENDIX B. CHANNEL GENERATION RULES

Decay channels are generated programmatically from the same discrete rules described in Section 4. Algorithm.

- (1) Form daughter pools: pseudoscalar (π, K, η) , vector (ρ, ω, ϕ) , open charm $(D, D_s, \Lambda_c, \dots)$, open bottom $(B, \Lambda_b, \dots)$, lepton pairs $(e^+e^-, \mu^+\mu^-)$.
- (2) Enumerate combinations consistent with quantum-number tags on the parent.
- (3) Discard kinematically closed channels ($Q \leq 0$).
- (4) Assign ledger tag and partial-width prior (OZI, CKM slot, $Q^{3/2}$ phase space for hadronic modes).
- (5) Normalize to branching ratios per parent.

Representative open-channel counts. at LHC $\sqrt{s} = 13$ TeV:

Parent	Generated	Minimum benchmark
J/ψ	190	100
$\Upsilon(1S)$	222	150
D^+	6	5
Λ_c	6	5

Light-hadron parents retain small static topologies; heavy-flavour and quarkonium parents use full expansion.

APPENDIX C. BENCHMARK PANELS

Table 4 summarizes structural comparison checks (masses, topology, facility access). The primary branching artifact is Table 2 in Section 7, backed by 490 open-channel readout rows in `data/hep_decay_benchmark.json`. Reference values follow PDG 2024 [20] and published facility kinematics [2, 4]; they never enter prediction formulas.

Mass tolerances default to 8% relative or 15 MeV absolute; proton and neutron use tighter bands. Branching tolerances are channel-specific (absolute or relative as declared). Half-life panel uses $\log_{10} |\text{pred}/\text{ref}| \leq 2$ envelopes.

APPENDIX D. REPRODUCIBILITY

Predictions, comparison rows, and pass/fail status export from the HQIV-LEAN repository. A short reproducibility index is maintained at `papers/hep_decay_readout/scripts/README.md`.

TABLE 4. Structural comparison checks (declared tolerances).

Panel	Cases
Mass	17
Kinematics	3
Decay topology (Q , openness, daughters)	37
Branching structural checks	5
Half-life (log envelope)	4
Environment	2
Production access	7
Production rate ordering	4
Multichannel minima	4
Branching diagnostic table	13
Open-channel readout (diagnostic)	490
Structural total	83

Entry points include the HEP decay readout mirror, multichannel expansion, production readout, uncertainty propagation, benchmark discharge, and unified decay-calculator CLI. Strict benchmark refresh and Lean build commands are documented there. Comparison inputs live in `data/hep_decay_observations.json`; outputs in `data/hep_decay_benchmark.json`.

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